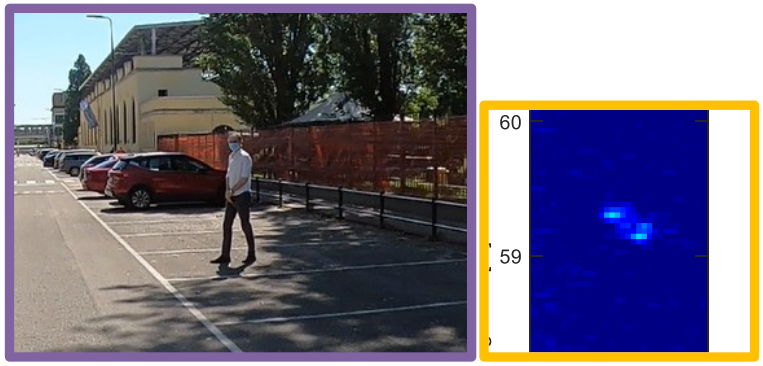
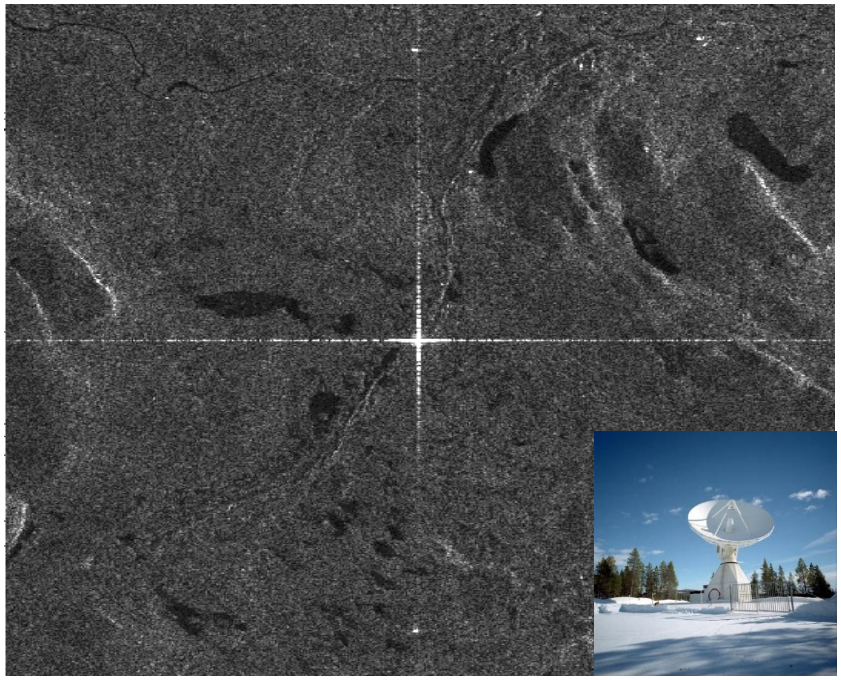
Targets, Radar Cross Section and the Radar Equation





Abstract

Received power: the radar equation (bistatic and monostatic)

Radar Cross Section

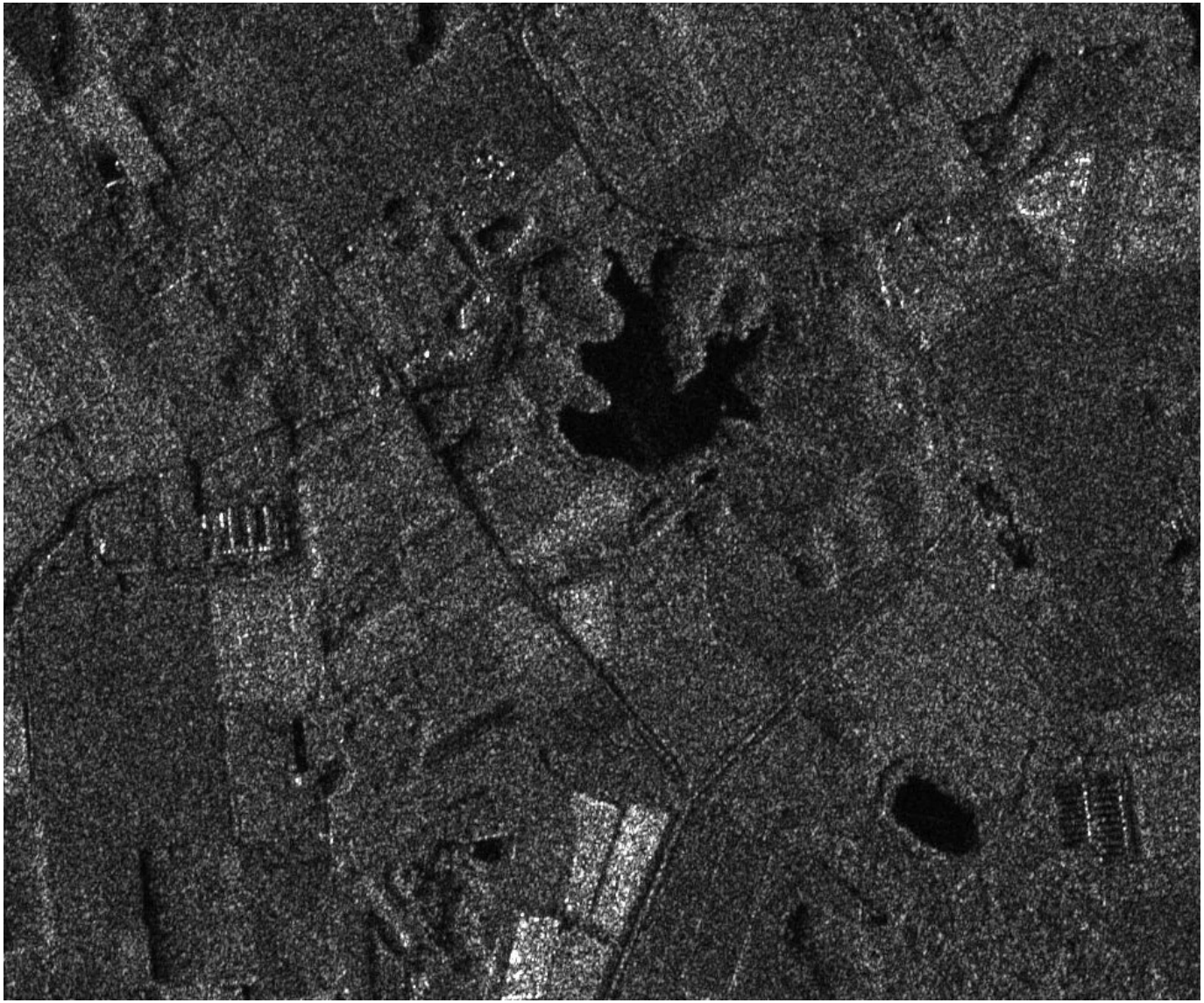
Point targets: RCS calculation

Distributed targets and the backscatter coefficient

Radiometric resolution and Equivalent Number of Looks (ENL)

Why does a person appear as a bright point whereas the surrounding terrain is grainy?

Point & distributed targets



Gaofen-3:
Image
magnitude

Identify and explain:

1) Bright targets (point scatterers)
2) Speckled, distributed targets?
3) Smooth surfaces

Point & distributed targets



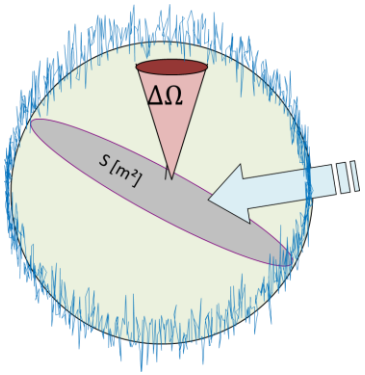
Example of high-resolution Ka-band SAR images, (4 inch ground resolution)

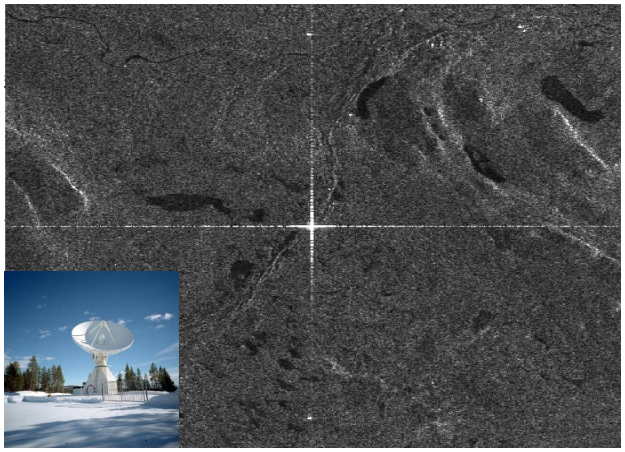
Sandia National Laboratories

Which image features reveal material properties, and which reveal scattering geometry?

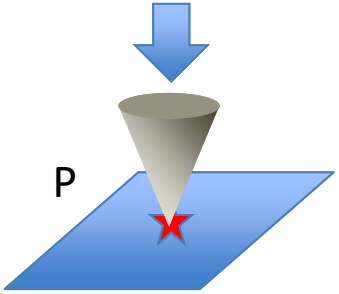
RCS: definition

1) Isotropic scatterer
2) Perfectly conducting rectangular plate, a x b, at normal incidence





RCS and directivity: Point targets



Mirror

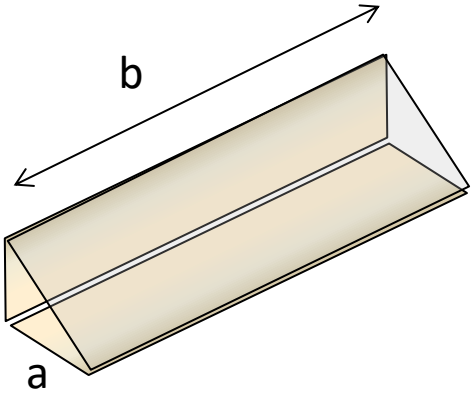
Example



RCS

Angular aperture

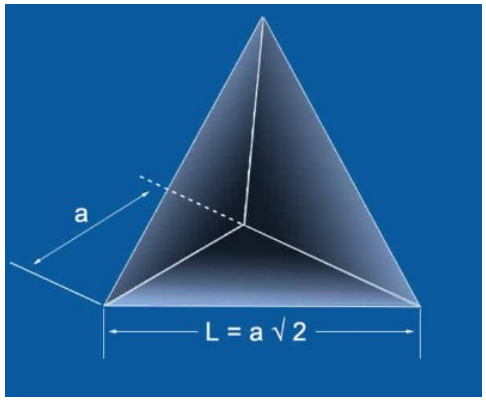
Phase center

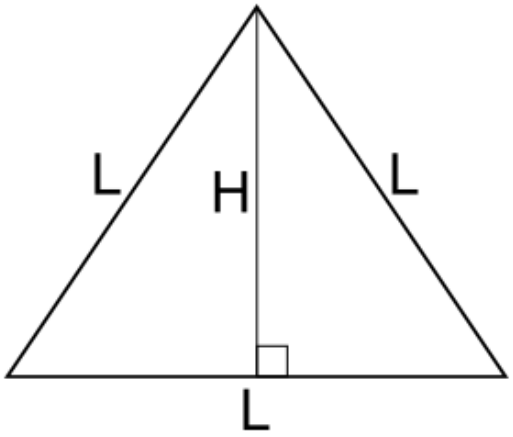


Dihedral



Trihedral RCS





$$A_{geo} = 2 \cdot \frac{\left(\frac{L}{2} \cdot H\right)}{2} = \frac{L^2\sqrt{3}}{4}$$

$$\sigma = \pi \frac{3}{4} \frac{L^4}{\lambda^2}$$

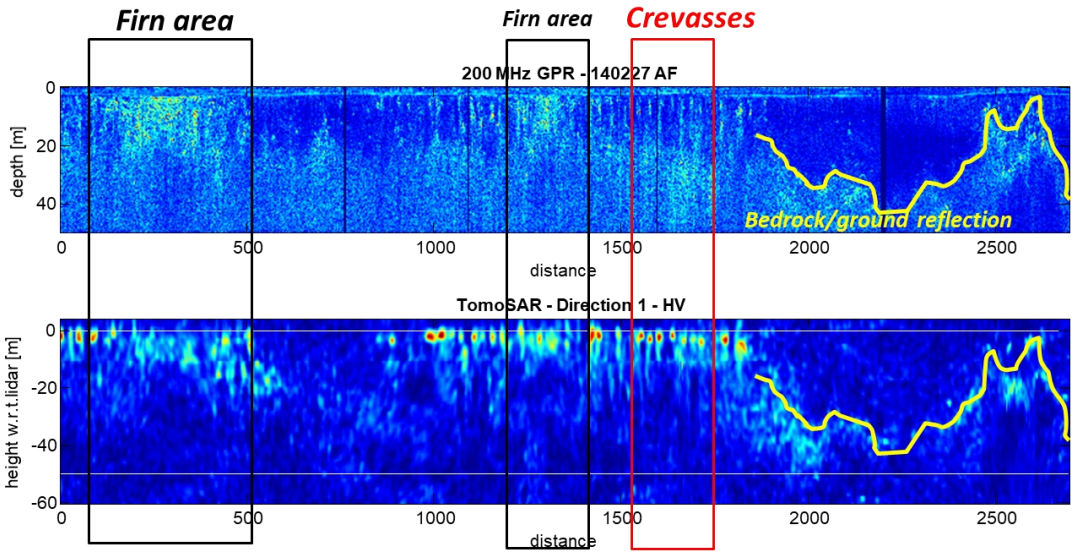
Quiz: trihedral RCS



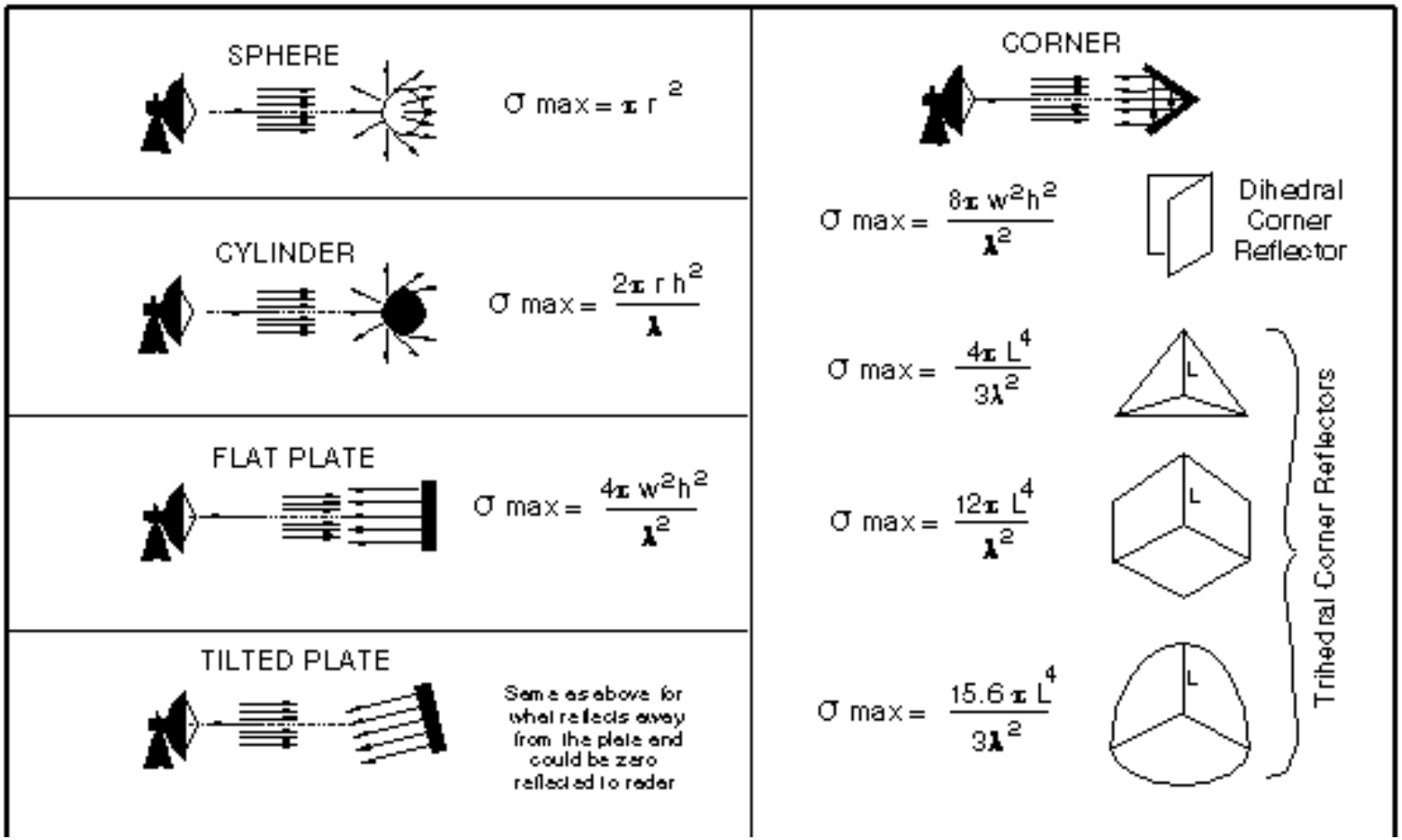
Aresys trihedral reflectors can be dismounted and carried manually for field campaigns.
The one here has L = 50 cm, was used for the Mittelbergferner airborne tomographic campaign.

Compute the RCS in dBsm at L and X band





RCS and directivity: Point targets



Example of real target RCS

Bee Model in WIPL-D

WIPL-D model of a bee is shown in Figure 1. EM simulation was performed @ 10 GHz. The model originates from a CAD file. The bee body is considered to be dielectric, with relative dielectric constant εr=30. Losses were not included.

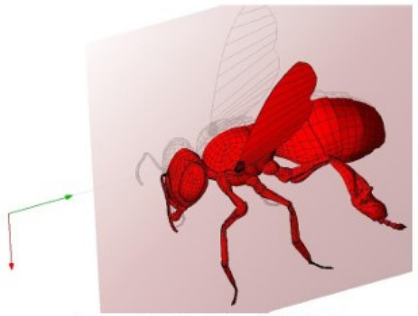


Figure 1. Model of a bee in WIPL-D.

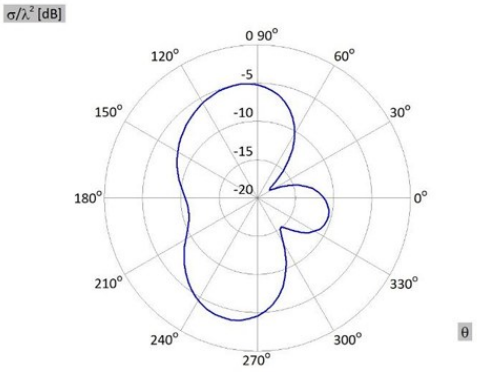


Figure 2. Monostatic RCS of the full 3D bee model.

Compute RCS at X band

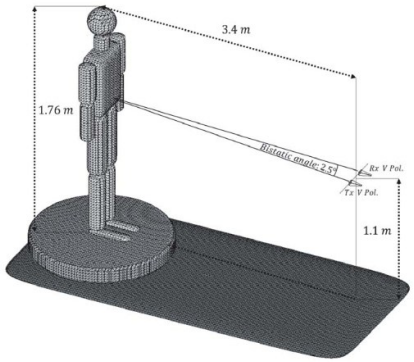
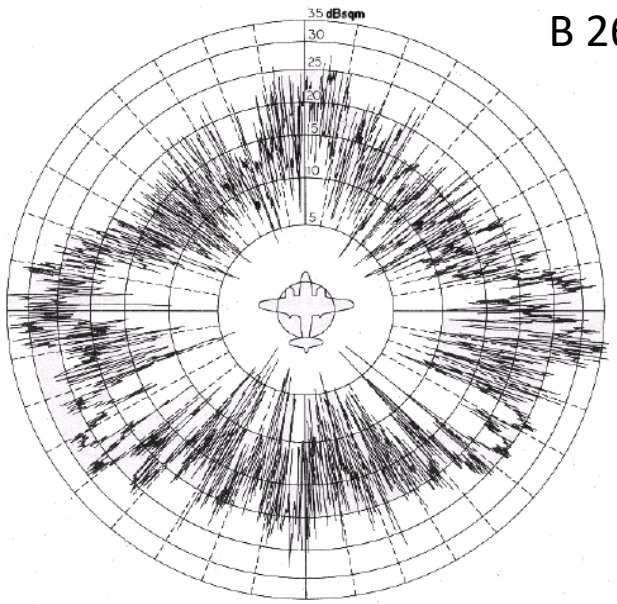


Figure 3. Setup for the evaluation of the RCS of a human target, reproduced in MATLAB® software.

Table 5. Simulated and measured average values of the the pedestrian look angles.

Look angles	Measured [15] (dBsm)	Simulated (dBsm)
Front	−5.47	−5.26
Rear	−2.19	−5.49
Right side	−3.90	−5.53
Left side	−4.97	−5.36

Example of real target RCS

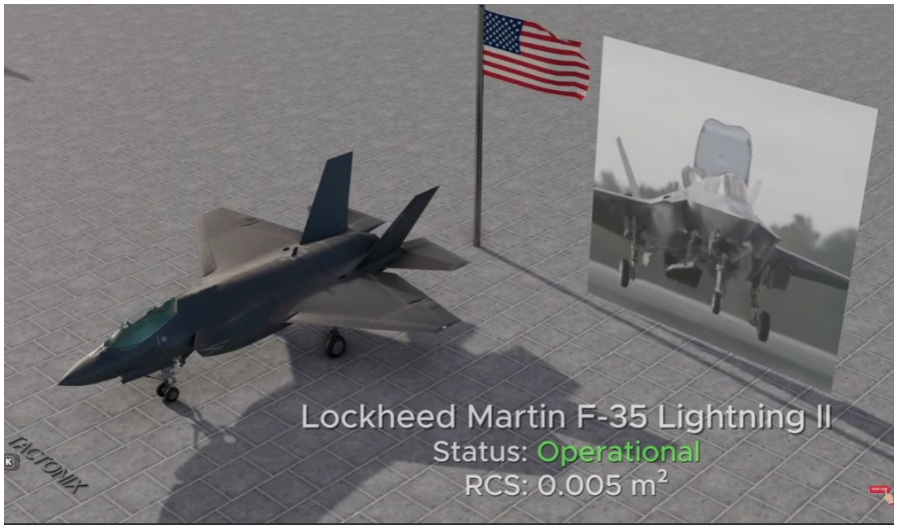


B 26 Bomber



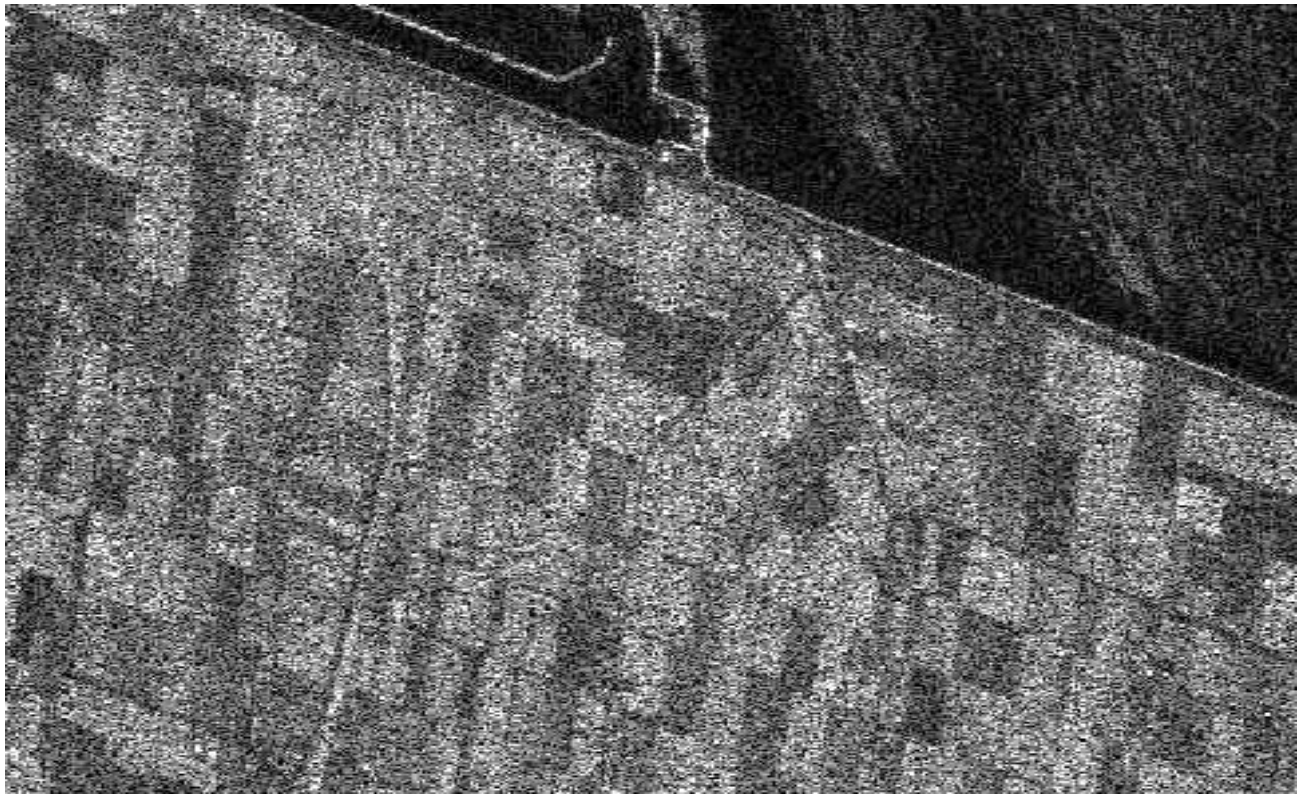
U. S. Bomber
B-26
"Marauder"







Distributed scatterer: RCS & backscatter coefficient



Within a homogeneous resolution cell, the complex scattering amplitude is approximately circular complex Gaussian.

Distributed target: RCS



Brownian motion (random walk)



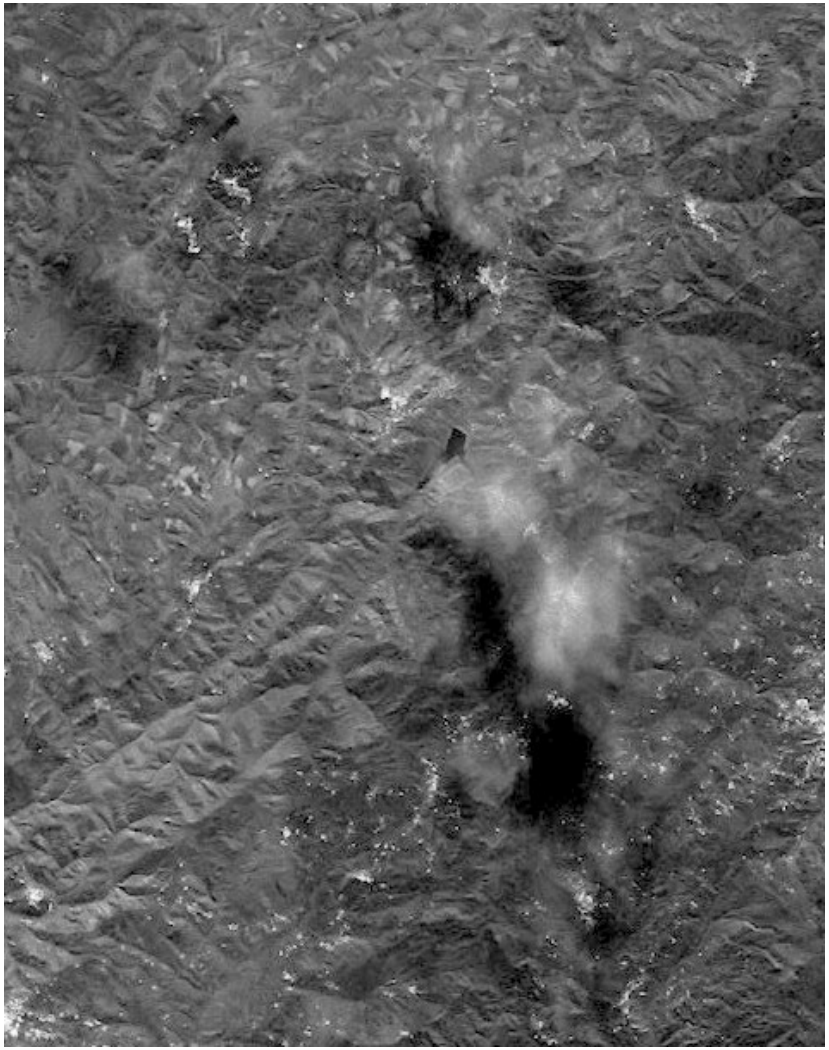
Volume/surface: backscatter coefficient

Surface scattering

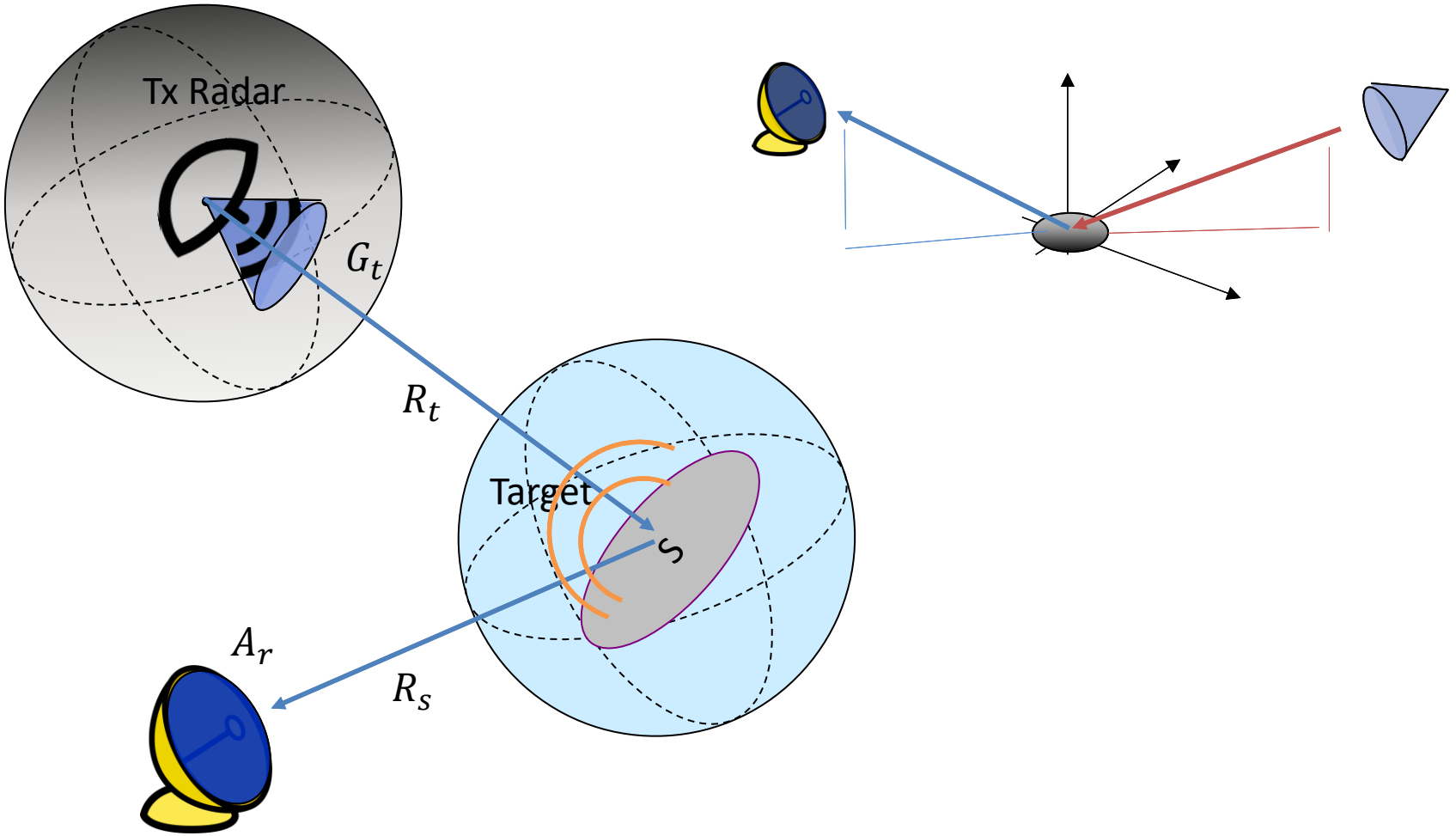
$$RCS\ [m^2] = \sigma^o\ S \rightarrow \sigma^o = RCS/S\ \ [m^2/m^2]$$

Volume scattering

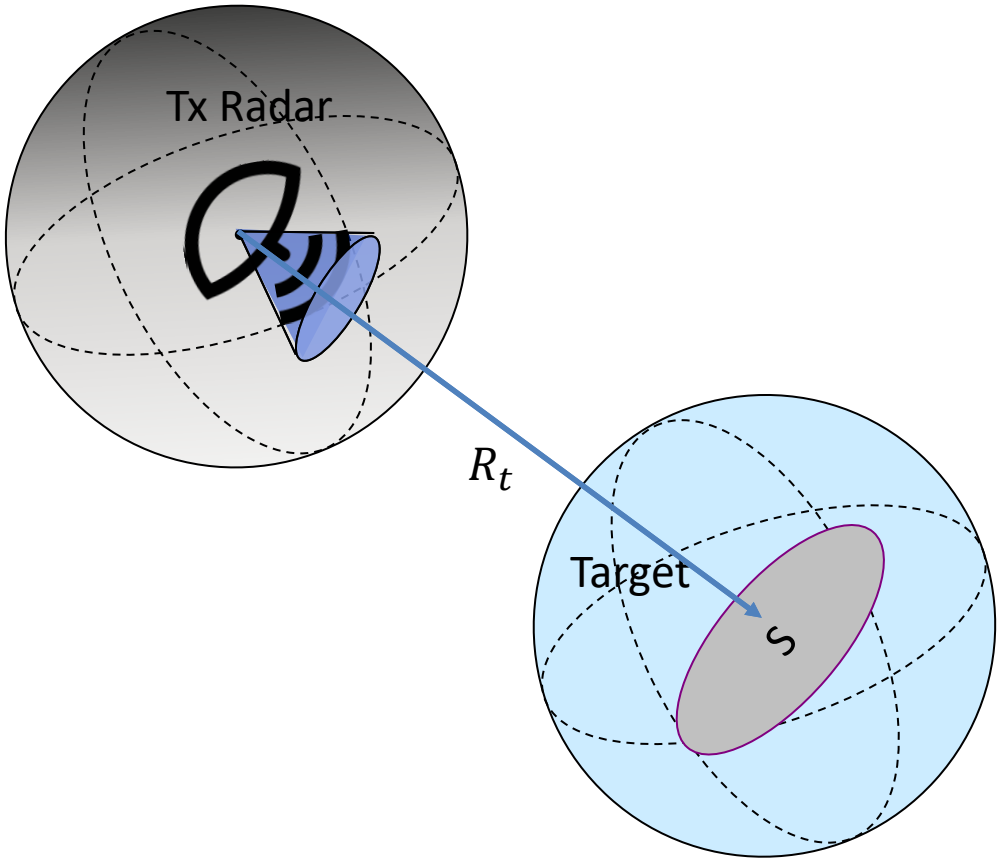
$$RCS\ [m^2] = \sigma^o\ V \rightarrow \sigma^o = RCS/S\ \ [m^2/m^3]$$



Compute the received power (Radar equation): bistatic case

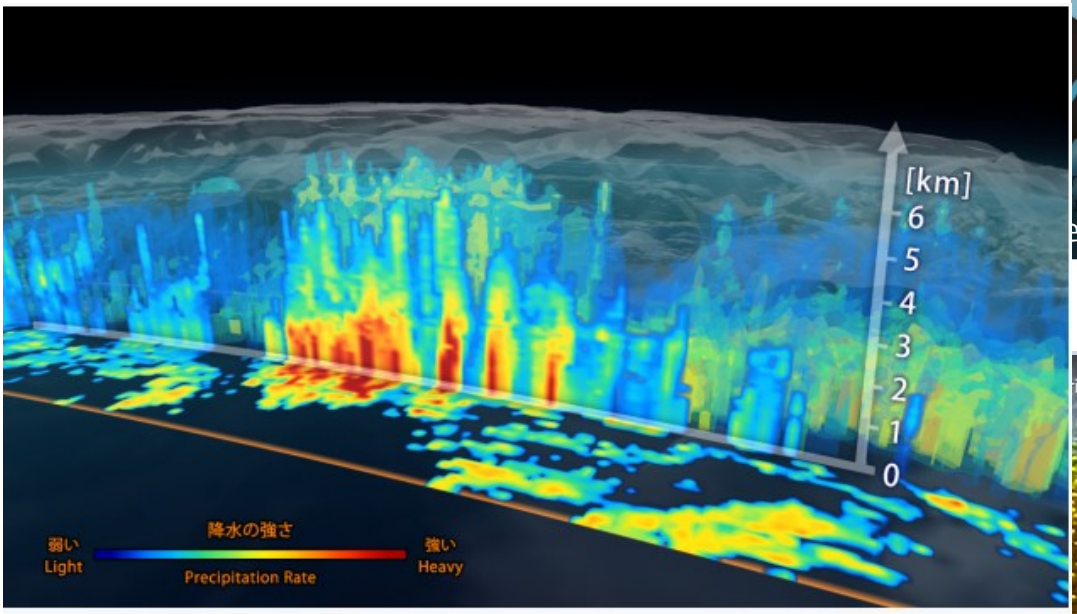


The Radar equation: monostatic case



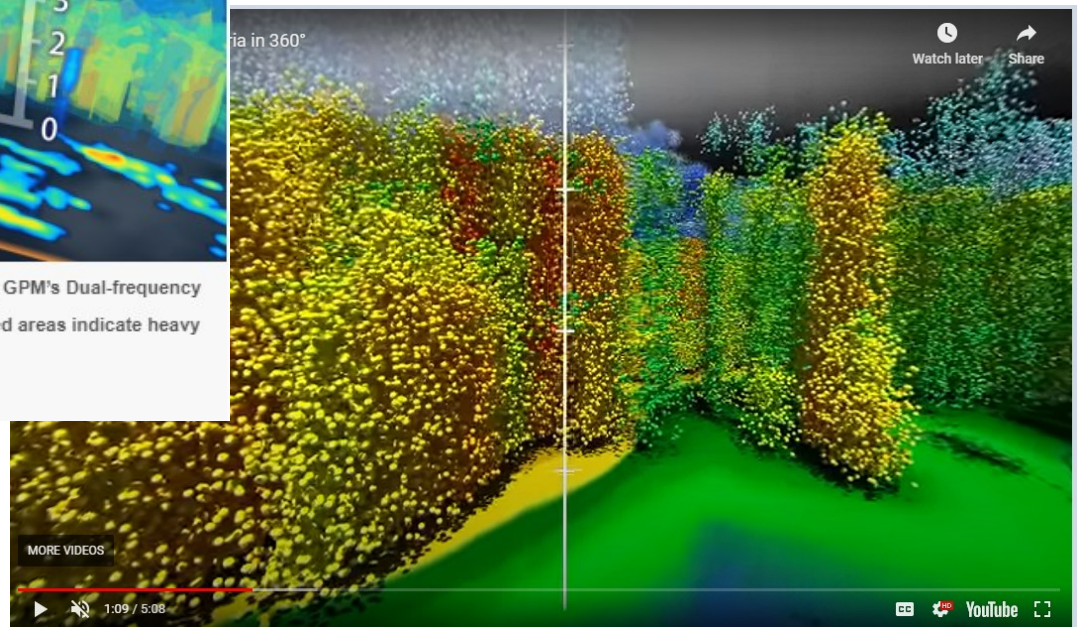
Volume scattering: meteorological Radar





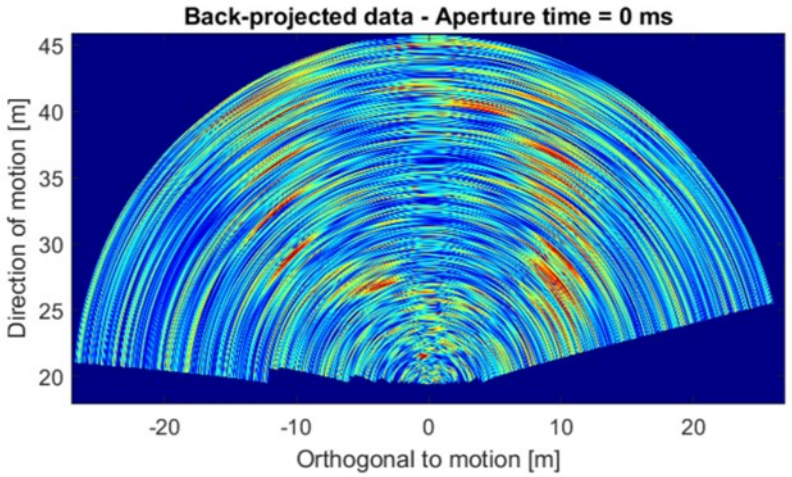
3D view inside an extra-tropical cyclone observed off the coast of Japan, March 10, 2014, by GPM's Dual-frequency Precipitation Radar. The vertical cross-section approx. 4.4 mi (7 km) high show rain rates: red areas indicate heavy rainfall while yellow and blue indicate less intense rainfall.

Image Credit: JAXA/NASA. Video here.



Example: automotive Radar







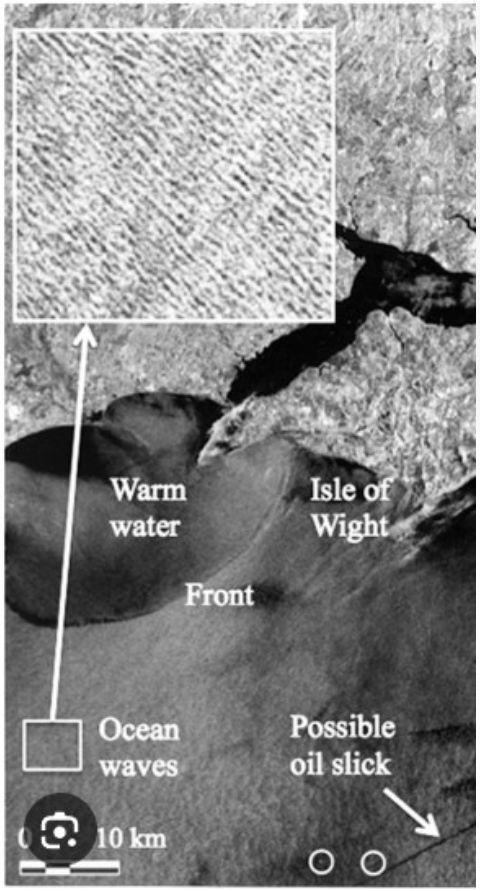
Compute the required power to achieve SNR > 15 dB from a target -5 dBsqm

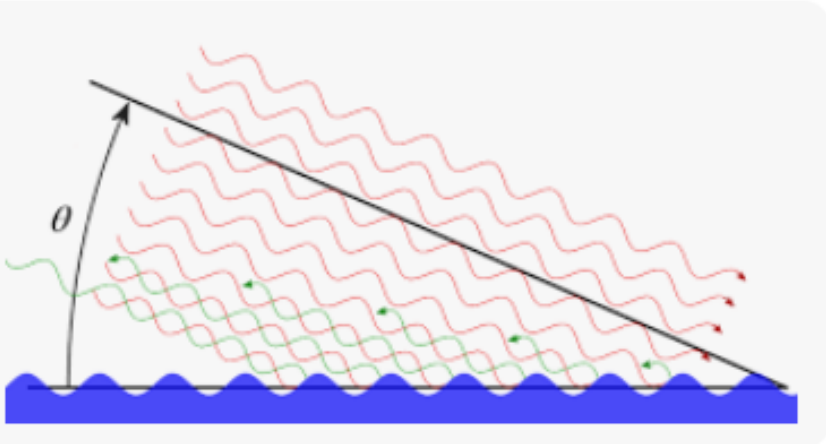
Assume:

- antenna aperture 20° (both elev and azimuth)
- Total losses 4 dB
- Range 100 m
- Noise factor of receiver 3 dB
- Observation time: such that a target at 100 km/h does not move more than 0.1 m in Tobs
- Duty cycle of tx 10%
- λ= 4 mm

A periodic grating: a Bragg scatterer







Compute the RCS as function of Δx, λ and θ

